

附: 积分公式的证明.

$$\text{proof } E[X] = \int_0^{\infty} x f(x) dx$$

$$\text{令 } x = \int_0^x 1 dt = \int_0^{\infty} I_{\{t \leq x\}} dt$$

$$E[X] = \int_0^{\infty} x f(x) dx = \int_0^{\infty} \left(\int_0^{\infty} I_{\{t \leq x\}} dt \right) f(x) dx$$

$$\text{Fubini-Tonelli} \Rightarrow = \int_0^{\infty} \int_0^{\infty} I_{\{t \leq x\}} f(x) dx dt$$

$$= \int_0^{\infty} \left(\int_0^{\infty} I_{\{t \leq x\}} f(x) dx \right) dt$$

$$= \int_0^{\infty} \left(\int_t^{\infty} f(x) dx \right) dt$$

$$= \int_0^{\infty} P(X > t) dt$$